GRAPHIC ERA (DEEMED TO BE UNIVERSITY), DEHRADUN

SEMESTER II

Name of Department: - Computer Science and Engineering

1. Subject Code: **PCS253** Course Title: **AI and ML Lab using Python**
2. Contact Hours: **L:** **0** **T**: **0** **P**: **2**
3. Examination Duration (Hrs): **Theory** **0** **Practical** **2**
4. Relative Weight: **CIE** **25** **MSE** **25** **ESE** **50**
5. Credits: **1**
6. Semester: **II**
7. Category of Course: **DSC**
8. Pre-requisite: Python Programming Lab (PCS152).

9. **Course Outcome:**	After completion of the course, the students will be able to**:** CO1: Design and develop programs to visualize data and perform Exploratory data analysis. CO2: Design and develop graph search algorithms such as BFS, DFS, and A* using heuristics. CO3: Design and develop Machine Learning models (Linear Regression, Logistic Regression, KNN, Decision Tree). CO4: Analyze and evaluate machine learning models using metrics (Confusion Matrix, F1 Score, Precision, Recall, Accuracy), clustering, and association rule mining.

10. **Details of the Course:**

Sl. No.	**List of problems for which students should develop programs and execute in the Laboratory**	**Contact Hours**
1.	**Exploratory Data Analysis and Visualization:** Implement a python program to create a dataset having columns student_id, student_name, age, height, weight, semester, gender, city and perform the following: i. Find the shape of the dataset. ii. Display the names of the columns. iii. Compute the no. of null values in each column. iv. Find unique names. v. Plot line graph between height and weight, height and age and weight and age. vi. Visualize the outliers present in height column using scatterplot. vii. Show the outlier in each column using boxplot. viii. Remove the outlier from the height column using iqr method.	2

	ix. Remove outlier from weight column using z-score method.	
2.	Develop a python program to create dataset having height, weight, gender column and perform the following: i. Compute BMI using dataframe operations. ii. Save the dataset into CSV file. iii. Import the csv file and load it into a dataframe. iv. Find shape and size of dataframe. v. Find the count of null values in Gender column. vi. Remove null from gender column. vii. Display the structural and statistical information of the dataset. viii. Create a histogram on gender column. ix. Draw a suitable plot on the data and deduce some inference. x. Perform aggregate functions on height, weight and BMI column. xi. Find the entries where BMI is less than 20	2
3.	Given two datasets(medical_records.csv and patient_details.csv), perform the following tasks: i. Merge both the datasets on patient_id column and using inner join. ii. Impute null values in name column with value “unknown”. iii. Rectify the negative values present in age column. iv. Make the values in gender column consistent by converting values M to Male and F to Female or vice-versa. v. Make the case of values in city column consistent. vi. Make the date format of admission_date consistent using regular expression. vii. Remove string value from height_cm column and impute the values. viii. Replace “high” from bp column with 140/90. ix. Impute sugar_level with mean value. x. Perform group by operation on gender column and compute mean, min and max of visit_cost for each gender. xi. Plot a histogram on diagnosis column.	2
4.	**AI Overview and Search:** Design and develop programs to implement the Depth-First Search (DFS) algorithm for graph traversal in Artificial Intelligence with the following specifications: i. Construct a static graph consisting of 6 nodes and 10 edges, where the graph is stored using a dictionary representation. The graph is given as: A- B, A-C, B-D, B-E, C-D, C-F, D-E, D-F, E-F, B-F. ii. Create another graph where the number of nodes and number of edges are provided by the user at runtime. iii. Accept the initial node and the goal node from the user. iv. Display the path from the initial node to the goal node for both the cases.	2
5.	Design and develop programs to implement the Breadth-First Search (BFS)	2

algorithm for graph traversal in Artificial Intelligence with the following specifications:

i. Construct a static graph consisting of 6 nodes and 10 edges, where the graph is stored using a dictionary representation.
The graph is given as:
A–B, A–C, B–D, B–E, C–D, C–F, D–E, D–F, E–F, B–F.
ii. Create another graph dynamically, where the number of nodes and number of edges are provided by the user at runtime, and the user specifies which nodes are connected.
iii. Accept the initial node and the goal node from the user.
iv. Perform Breadth-First Search (BFS) traversal starting from the given initial node and:
v. Display the order of node traversal
vi. Display the shortest path from the initial node to the goal node for both the static and dynamic graph cases.

6. Construct a program to implement the A* search algorithm using heuristic values for a given graph. The graph is stored using a dictionary representation. The graph consists of the following nodes and edge costs: 2

Path	**Edge Cost**
A-B	1
A-C	3
B-D	3
B-E	6
C-D	1
D-E	1

Node	A	B	C	D	E
h(n)	7	6	4	2	0

Accept the initial node from the user and node E as goal node. Apply the A* algorithm to:

i. Display the order of node expansion
ii. Display the optimal path
iii. Display the total path cost

7. **Machine Learning and Preprocessing:** 2

Develop a Python program to perform normalization on a given healthcare dataset using Min–Max Scaler and Standard Scaler. The program should perform the following tasks:

i. Load the given dataset from the csv file(healthcare.csv).
ii. Apply Min–Max normalization to scale the features.
iii. Apply Standard Scaler to the same features.
iv. Plot appropriate graphs to visualize the normalized data obtained from both methods.
v. Plot appropriate graphs to visualize the normalized data obtained from both methods.

8. Build a Python program to implement Forward Feature Selection and 2

	Backward Feature Elimination techniques using a given healthcare dataset that contains 200 records with multiple patient-related features. The program should perform the following: i. Load the dataset into dataframe. ii. Apply Forward Feature Selection using a Linear regression, kNN models and compare the results. iii. Apply Backward Feature Elimination using the same machine learning models and analyze the results. iv. Compare the feature sets selected by Forward Feature Selection and Backward Feature Elimination.	
9.	Implement a program to perform dimensionality reduction using PCA on a Smart City Monitoring dataset(smart_city_monitor.csv) containing 100 samples and 20 features. The program should perform the following tasks: i. Load the given dataset from a CSV file. ii. Standardize the dataset before applying PCA. iii. Apply Principal Component Analysis (PCA) to reduce the dimensionality of the dataset. iv. Determine the number of principal components required to retain a significant amount of variance. v. Transform the dataset into the reduced feature space. vi. Visualize the data before and after dimensionality reduction and analyze the results.	2
10.	**Regression Analysis** You are given a dataset (salry_prediction.csv) containing years_of_experience as an input feature and salary as target. Implement a program to perform Simple Linear Regression and analyze the relationship between an input feature and a target variable and perform the following: i. Read the dataset into a dataframe and train a simple linear regression model after splitting the dataset into train and test data in the ratio of 70:30. ii. Determine the slope and intercept of the best fit line. iii. Plot the regression line along with actual data points. iv. Use the trained model to predict salary of new data points. v. Evaluate the performance of the model using appropriate regression metrics.	2
11.	Consider a dataset (salary_prediction2.csv) containing multiple input features related to employee performance and a target variable Salary. Implement a program to perform Multiple Linear Regression and analyze the relationship between multiple input features and the target variable. Perform the following tasks: i. Read the given dataset into a dataframe. **ii.** Split the dataset into training and testing sets in the ratio 70:30, 60:40, 80:20**.** iii. Train a Multiple Linear Regression model using all input features to predict the target variable Salary. iv. Determine the coefficients of each input feature and the intercept of the best-fit regression model.	2

	v. Plot appropriate graphs to compare actual vs predicted salary values. vi. Use the trained model to predict the salary for new data points taken as input from the user. vii. Evaluate the performance of the model using appropriate regression metrics.	
12.	**Classification** You are working on a dataset(diesea_risk.csv) containing multiple input features related to patient health parameters and a binary target variable Disease_Risk. Implement a program to perform Logistic Regression for binary classification and analyze the relationship between the input features and the target variable. Perform the following tasks: i. Read the given dataset into a dataframe. ii. Split the dataset into training and testing sets in the ratio 70:30. iii. Train a Logistic Regression model using the input features to predict the binary target variable Disease_Risk. iv. Determine the model coefficients and intercept. v. Evaluate the performance of the model using accuracy, and show the confusion matrix. vi. Use the trained model to predict the class label for new data points.	2
13.	Develop a program to implement the K-Nearest Neighbours (KNN) algorithm on a given dataset related to student performance with target as Result(student_performance.csv). The program should perform the following tasks: i. Read the given dataset into a dataframe. ii. Separate the input features and the target class label (`Result`). iii. Split the dataset into training and testing sets in the ratios 70:30, 60:40, 80:20. iv. Implement the K-Nearest Neighbours (KNN) classification algorithm for a suitable value of K. v. Train the model using the training data and classify the test samples. vi. Evaluate the performance of the model using accuracy, precision, F1 score and recall and show the confusion matrix. vii. Use the trained model to predict the class label for new data points.	2
14.	Consider a dataset(loan_approval_dataset.csv) containing records related to loan applicants, where multiple input features are provided along with a binary target class label **`Loan_Approval`**. The program should perform the following tasks: i. Read the given dataset into a dataframe. ii. Separate the input features and the target class label (`Loan_Approval`). iii. Split the dataset into training and testing sets in the ratio 70:30. iv. Train a Decision Tree Classifier using the training data. v. Use the trained model to classify the test data. vi. Evaluate the performance of the classifier using appropriate classification metrics. vii. Predict the class label for new loan applicant data points.	2
15.	**Clustering and Association Rule Mining:**	2

	Consider a dataset containing records related to customer purchasing behavior. The dataset includes multiple numerical features describing customer demographics and spending patterns.The program should perform the following tasks: i. Read the given dataset into a dataframe. ii. Perform necessary data preprocessing and feature scaling. iii. Apply the K-Means Clustering algorithm on the dataset for different values of K. iv. Determine the optimal number of clusters using the Elbow Method. v. Validate the clustering results using the Silhouette Score. vi. Perform clustering using the optimal value of K and assign cluster labels to the data points. vii. Visualize the formed clusters and analyze the clustering results.	
16.	You are provided with a transactional market basket dataset (market_basket_dataset.csv), where each transaction contains a set of items purchased together by a customer. Develop a program to demonstrate Association Rule Mining using the Apriori Algorithm. The program should perform the following tasks: i. Read the transactional dataset from a CSV file. ii. Convert the dataset into a suitable format for association rule mining. iii. Apply the Apriori algorithm to generate frequent itemsets based on a minimum support threshold. iv. Generate association rules from the frequent itemsets using minimum confidence. v. Display the discovered frequent itemsets and association rules along with their support, confidence values.	2
	Total	**32**

Text Books:

Authors Name	Title	Edition	Publisher, Country	Year
Stuart Russell, Peter Norvig	Artificial Intelligence: A Modern Approach	4th Edition	Pearson, USA	2021
Ethem Alpaydin	Introduction to Machine Learning	4th Edition	MIT Press, USA	2020
Aurélien Géron	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow	3rd Edition	O'Reilly Media, USA	2022

Reference Books:

Authors Name	Title	Edition	Publisher, Country	Year
Tom M. Mitchell	Machine Learning	1st Edition	McGraw-Hill, USA	1997
Yahwant Kanetkar	Let Us Python	5th Edition BPB,	India	2023

Course Articulation Matrix

CO	Statement	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
PCS253.1	Design and develop programs to visualize data and perform Exploratory data analysis.	3	3	2	3	3	1	1	1	1	1	1	3	3	1
PCS253.2	Design and develop graph search algorithms such as BFS, DFS, and A* using heuristics.	3	2	2	2	3	-	1	-	-	1	-	3	2	1
PCS253.3	Design and develop Machine Learning models (Linear Regression, Logistic Regression, KNN, Decision Tree).	3	3	2	2	3	-	2	-	-	2	-	3	3	2
PCS253.4	Analyze and evaluate machine learning models using metrics (Confusion Matrix, F1 Score), clustering, and association rule mining.	3	3	2	3	3	-	2	-	-	2	-	3	3	3
PCS253		3	2.75	2	2.5	3	1	1.5	1	1	1.5	1	3	2.75	1.75

High correlation (3); Medium correlation (2); Low correlation (1), No correlation (-)